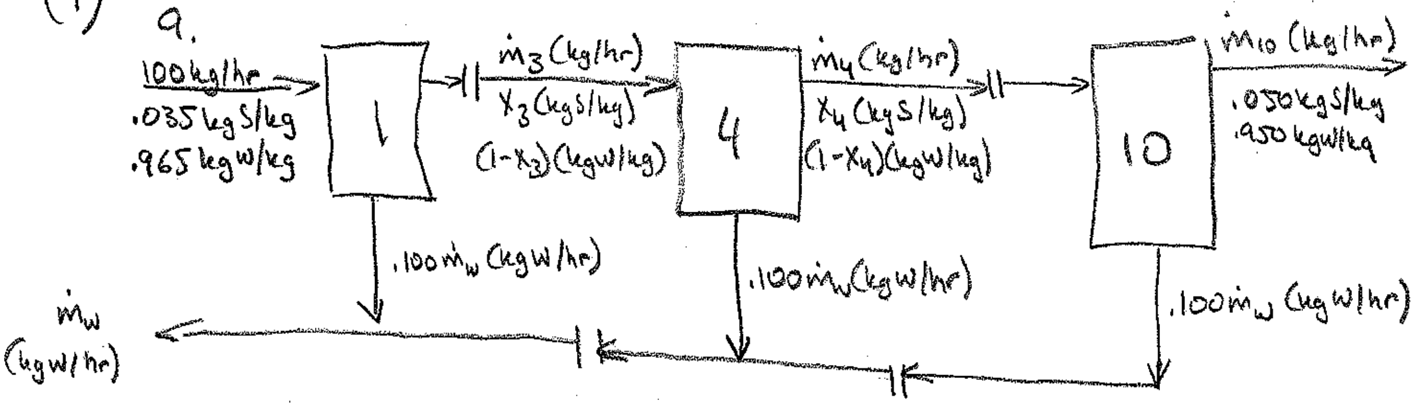
Homework #4 Solutions

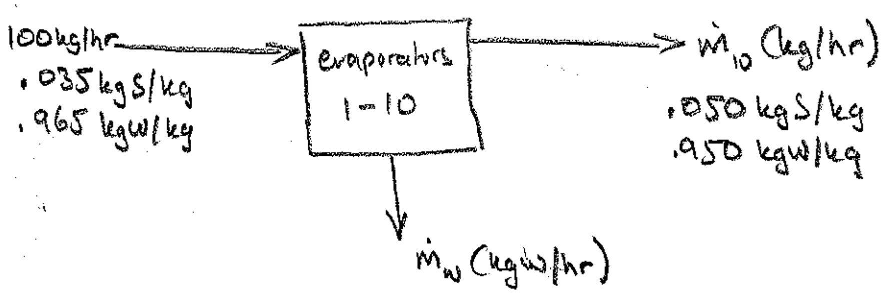
ChE 210, Fall '08
Yevgenny Strashnov
(TA)

(1)

a.



b. Overall process diagram:

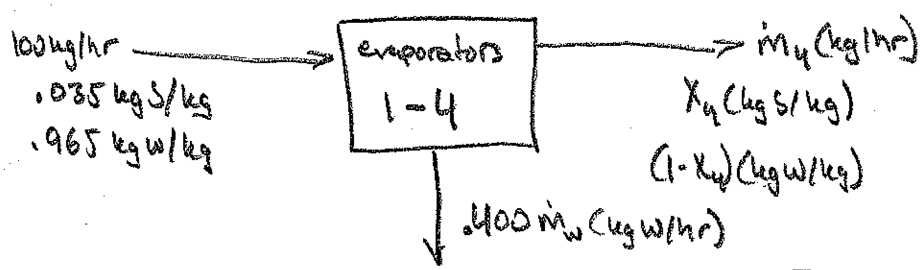


Salt Balance: $.035(100) = .050\,\dot{m}_{10}$

Overall Balance: $100 = \dot{m}_w + \dot{m}_{10}$

H_2O Yield: $Y_w = \dfrac{\dot{m}_w}{96.5}$ ← kg W recovered / ← kg W in feed

First Four Evaporators:



Overall Balance: $100 = .4\dot{m}_w + \dot{m}_4$

Salt Balance: $.035(100) = X_4 \dot{m}_4$

c.

Solving the equations from part b., we get:

$\dot{m}_{10} = 70$ kg/hr

$\dot{m}_w = 30$ kg/hr

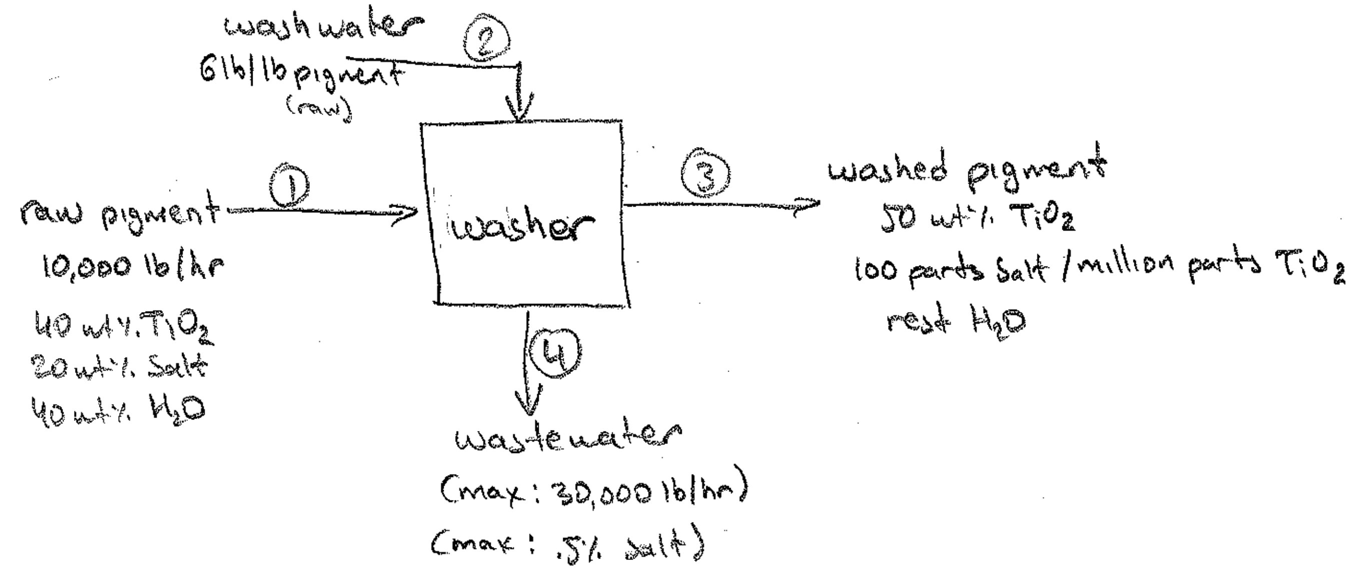
$\boxed{Y_w = .31}$

&

$\dot{m}_4 = 88$ kg/hr

$\boxed{X_4 = .0398}$

(2) Murphy 2.66



DOF Analysis: (for detailed procedure, see Murphy, pp. 104-111)

Variables		Constraints	
Stream:	9	Specified Flows:	2 (① & ④)
System:	0	Specified Compositions:	7 (3 in ①, 1 in ②, 2 in ③, & 1 in ④)
		System performance:	0
		Material Balances:	3
	9		12

$\therefore$ DOF $= 9 - 12$
$= -3$

The DOF Analysis shows that the problem is <u>much</u> overspecified... However, some of the constraints are "soft" in that they are maximums (or minimums) & optimums. They are not hard requirements...

In this situation, one approach is to ignore the soft constraints & solve for the other variables first as though the other constraints did not exist. Then, the solution can be checked against the other constraints and see if they are met...

③ The remaining part of the solution is optional and is shown purely for your learning benefit...

Temporarily, ignore the constraints that the problem statement imposes on wastewater quality & quantity; allow the constraints on salt concentration in the washed pigment & on the optimum washwater feed rate to remain in effect...

TiO_2: $X_{T,1} = .4(10{,}000) = 4000 \text{ lb/hr} = X_{T,3}$

Salt: $X_{S,1} = .2(10{,}000) = 2000 \text{ lb/hr} = X_{S,3} + X_{S,4}$

H_2O: $X_{w,1} + X'_{w,2} = .4(10{,}000) + 6(10{,}000) = 64000$

then, from the specs on stream ③ (washed pigment):

$$\frac{X_{S,3}}{X_{T,3}} = \frac{100}{1{,}000{,}000} \Rightarrow X_{S,3} = \frac{100}{1{,}000{,}000}(4000) = .4 \text{ lb/hr}$$

$$\& \quad \frac{X_{T,3}}{X_{T,3}+X_{S,3}+X_{w,3}} = \frac{4000}{4000+.4+X_{w,3}} = .5 \Rightarrow X_{w,3} = 3999.6 \text{ lb/hr}$$

then, from the mass balances,

$X_{S,4} = 2000 - .4 = 1999.6 \text{ lb/hr}$

$X_{w,4} = 64000 - 3999.6 = 60{,}000 \text{ lb/hr}$

$$\text{salt conc.} = \frac{1999.6}{60000} \times 100\% = 3.33\%$$

So we have double the allowed water dump volume and the salt concentration in it is more than 6 times the maximum allowed by the spec!

In order to meet all these specifications, the process will need to be re-engineered! :)

④

(3) Murphy 2.69 (solve the equations)

DOF Analysis:

	Variables	
Stream	25	- 3 in str. 1 3 in str. 2 5 in " 3 3 in 4 2 in 5 1 in 6 1 in 7 1 in 8 2 in 9 2 in 10 2 in 11 25
System	2	- 2 chemical rxns., steady-state ∴ no accum.
tot:	27	

	Constraints	
Spec. Flows	1	- cumene production rate
Spec. Comps.	4	- 95/5 P/I in str. 1 1.2/1 P/B also in str. 1 10% P in str. 2 the split performed by the splitter
Spec. System Perf.	2	- 80% conversion P 72% conversion B

5

Material Balances: 20 — 3 for mixer

5 for reactor

5 for V/L separator

2 for splitter

3 for dist. column 1

2 for dist. column 2

20

total: $1+4+2+20=27$

$\therefore$ DOF $= 27-27=0$!

we have 2 chemical reactions:

$$(P)\ C_3H_6 + \overset{(B)}{C_6H_6} \rightarrow C_9H_{12}\ (C) \qquad ①$$

$$+ (I)\ C_3H_8$$

$$\underset{(P)}{C_3H_6} + \underset{(C)}{C_9H_{12}} \rightarrow \underset{(D)}{C_{12}H_{18}} \qquad ②$$

Equations:

Basis: $X_{C,6} = 25$ mol/s $\Rightarrow 1$

Rxn Stoich.: $\dfrac{C_{gen①}}{B_{con①}} = \dfrac{1}{1}$, $\dfrac{P_{con①}}{B_{con①}} = \dfrac{1}{1}$, $\dfrac{D_{gen②}}{C_{con②}} = \dfrac{1}{1}$, & $\dfrac{P_{con②}}{C_{con②}} = \dfrac{1}{1}$ $\Rightarrow 4$

Stream Comp.: $\dfrac{X_{P,1}}{X_{I,1}} = \dfrac{95}{5}$, $\dfrac{X_{P,1}}{X_{B,1}} = \dfrac{1.2}{1}$, & $\dfrac{X_{I,2}}{X_{I,2}+X_{P,2}+X_{B,2}} = .1$ $\Rightarrow 3$

⑥

Splitter: $\frac{X_{P,9}}{X_{I,9}} = \frac{X_{P,10}}{X_{I,10}}$ $\Rightarrow 1$

System performance: $P_{con①} + P_{con②} = .8\,X_{P,2}$

$\Rightarrow 2$

& $B_{con①} = .72 X_{B,2}$

Material Balances:

mixer: $\Rightarrow 3$

$X_{P,2} = X_{P,1} + X_{P,10}$

$X_{I,2} = X_{I,1} + X_{I,10}$

$X_{B,2} = X_{B,1} + X_{B,b}$

reactor: $\Rightarrow 5$

$X_{P,3} = X_{P,2} - P_{con①} - P_{con②}$

$X_{I,3} = X_{I,2}$

$X_{B,3} = X_{B,2} - B_{con①}$

$X_{C,3} = C_{gen①} - C_{con②}$

$X_{D,3} = D_{gen②}$

V/L separator: $\Rightarrow 5$

$X_{P,9} = X_{P,3}$

$X_{I,9} = X_{I,3}$

$X_{B,4} = X_{B,3}$

$X_{C,4} = X_{C,3}$

$X_{D,4} = X_{D,3}$

splitter: $\Rightarrow 2$

$X_{P,10} + X_{P,11} = X_{P,9}$

$X_{I,10} + X_{I,11} = X_{I,9}$

dist. col. 1: $\Rightarrow 3$

$X_{B,8} = X_{B,4}$

$X_{C,5} = X_{C,4}$

$X_{D,5} = X_{D,4}$

dist. col. 2: $\Rightarrow 2$

$X_{C,6} = X_{C,5}$

$X_{D,7} = X_{D,5}$

Solutions to the equations:

- simply typing in the equations & variables into any software eq. system solver (such as QuickMath Automatic Math Solutions, search in Google), spits out a couple of solutions:

Sol. 1

$C_{gen}① = B_{con}① = P_{con}① = 17.56$ mol/s

$D_{gen}② = C_{con}② = P_{con}② = -7.443$ mol/s ← this cannot be a negative #, since the reaction ② only runs forward.

:(

Sol. 2

gen/con ① $= 27.67$ mol/s ✓

gen/con ② $= 2.668$ mol/s ✓

another negative #!

$X_{P,1} = 33.2$
$X_{B,1} = 27.7$
$X_{I,1} = 1.75$
...
$X_{P,11} = 2.87$
$X_{I,11} = 1.75$

$X_{P,2} = 37.9$
$X_{B,2} = 3.71$
$X_{I,2} = 4.63$

$X_{P,3} = 7.58$
$X_{I,3} = 4.63$
$X_{B,3} = -23.96$
$X_{C,3} = 25$
$X_{D,3} = 2.67$

$X_{B,4} = -23.96$
$X_{C,4} = 25$
$X_{D,4} = 2.67$
...
$X_{B,8} = -23.96$

$X_{C,5} = 25$
$X_{D,5} = 2.67$
...
$X_{P,9} = 7.58$
$X_{I,9} = 4.63$

$X_{C,6} = 25$
...
$X_{D,7} = 2.67$
...
$X_{P,10} = 4.72$
$X_{I,10} = 2.88$

Aside from the negative value of B leaving the reactor which is of course impossible, this soln. seems to be mostly ok...
The reason for the negative value may be a problem mis-spec.!

:(